Welcome to Multi-Engine

Welcome to the multi-engine phase of your program. Before getting started there are a few things you need to accomplish to show up as prepared as possible for the fast paced transition into multi-engine aircraft.

Things to know day 1:

1. V Speeds
2. Engine failure flow - FULL, CLEAN, KILL
3. Commercial Maneuvers, Single Engine Approach Profile, Gear Down Before Landing Checklist
4. Complete all Multi-Engine GIS videos
5. How to calculate performance using the POH
6. Study the Piper Seminole Exterior Components Worksheet in foreflight
7. Review the Commercial ACS for all commercial (AMEL) tasks

Commercial Multi-Engine Training Outline:

Day 1: Check-In & Ground
Day 2: Sim & Ground
- V Speeds and Single Engine Flow must be memorized prior to the first sim***

Day 3: Sim & Flight
- ALL configurations and maneuvers must be memorized prior to first flight***

Day 4: Flight
Day 5: Flight
Day 6: Endorsement Flight & Ground
Day 7: Checkride

*****Failure to have required items memorized will be a cancel charge*****

To be ready for checkride/endorsement you should be able to:

- Complete weight and balance
- List V-Speeds from memory
- Explain the critical engine and why the left engine is critical on a conventional twin (PAST)
- Explain the Vmc table (SMACFUM)
- Explain the fuel system
- Explain the electrical system
- Explain the prop system/hub/governor and how a constant speed prop works
- Explain the gear system
- Explain the vacuum system
- Understand emergency operations with each system
- Explain the characteristics of a T tail vs Conventional tail aircraft

T tail vs Conventional:

Advantages:

- Reduced drag and increased stability:
  - Reduced interference drag from the propellers downwash and turbulent airflow generated by the wings and fuselage. This is most pronounced at higher airspeeds.
- Effective pitch control:
  - Due to the reduction in interference drag, the elevator receives “cleaner” airflow allowing for more effective pitch control at most airspeeds.
- Exhaust blast:
  - T tails are removed from the effects of exhaust blast on aircraft with fuselage mounted engines.

Disadvantages:

- Slow speed aerodynamics:
  - The elevator of a T tail requires more control pressure to raise the nose due to the lack of prop reaching the control surface. In Conventional, the prop wash exerts a downward force on the tail helping to raise the nose.
- Flutter:
  - T tails require an increase in weight and control stiffness in order to prevent flutter at higher airspeeds
- Deep Stalls:
  - With the tail being above the wake of the wings, it is harder to recognize the indications of an impending stall which will lead the pilot to increase back pressure placing the aircraft into a deep stall. A deep stall is a high AOA stall in which the wake from the stalled wings spreads above and below the T tail effectively “blanketing” the elevator with decelerated, turbulent air rendering the tail ineffective.



Figure 6-11. *Aircraft with a T-tail design at a high AOA and an aft CG.*

Critical Engine Factors (PAST):

Critical Engine Factors: P-Factor
Critical Engine Factors: Accelerated Slipstream
Critical Engine Factors: Spiraling Slipstream
Critical Engine Factors: Torque

Single Engine Performance:

Loss of Climb Performance on a Single Engine

Vmc table videos (SMACFUM):

Vmc Basics
The VMC Table - Part 1 (Table Layout)
The VMC Table - Part 2 (Center of Gravity)
The VMC Table - Part 3 (Bank and Weight)
The VMC Table - Part 4 (Power and Density Altitude)
The VMC Table - Part 5 (Drag Devices and Feathering)
The VMC Table - Part 6 (CG and Performance)
The VMC Table - Part 7 (Bank and Performance)
The VMC Table - Part 8 (Performance with drag devices, power, weight, and prop condition)

Piper Seminole PA-44 Systems:

Drawing the PA-44 Vacuum System
Drawing the PA-44 Gear System
Drawing the PA-44 Fuel System Diagram
Drawing the PA-44 Prop Diagram
The Prop Governor: The Prop Hub
The Prop Governor: Flyweights
Drawing the PA-44 Electrical System

Additional Resources / Required Reading:

Read: FAA: Flying Light Twins Safely
Read the POH: Section 3, 4 (W&B), 5 (Performance), and 7

MEMORIZE ALL DRAWINGS FOR THE SYSTEMS
This is exactly how they need to be drawn out for your checkride

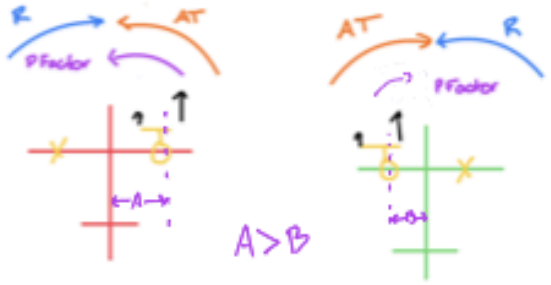
PAST

Conventional Twin

P-Factor

Climb

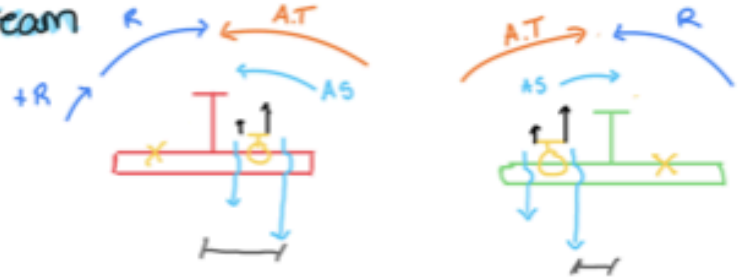
Yaw



Accelerated Slip Stream

Climb

Roll

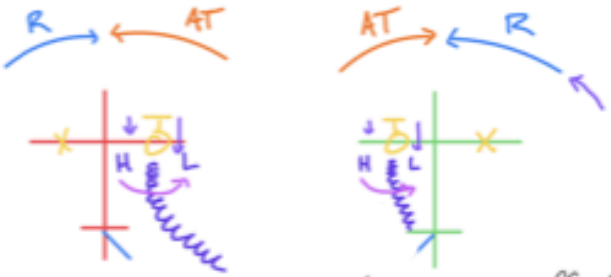


Due to Prop wash

The greater the amount of airflow over the wing is generating more lift over that segment of the wing. Therefor the banking tendency will be greater when the L critical engine fails.

Spiraling Slipstream

Yaw



* Have more effective rudder w/ airflow allowing the rudder to oppose Asymetric thrust

Torque

Roll



Torque will contribute to Asymetric thrust

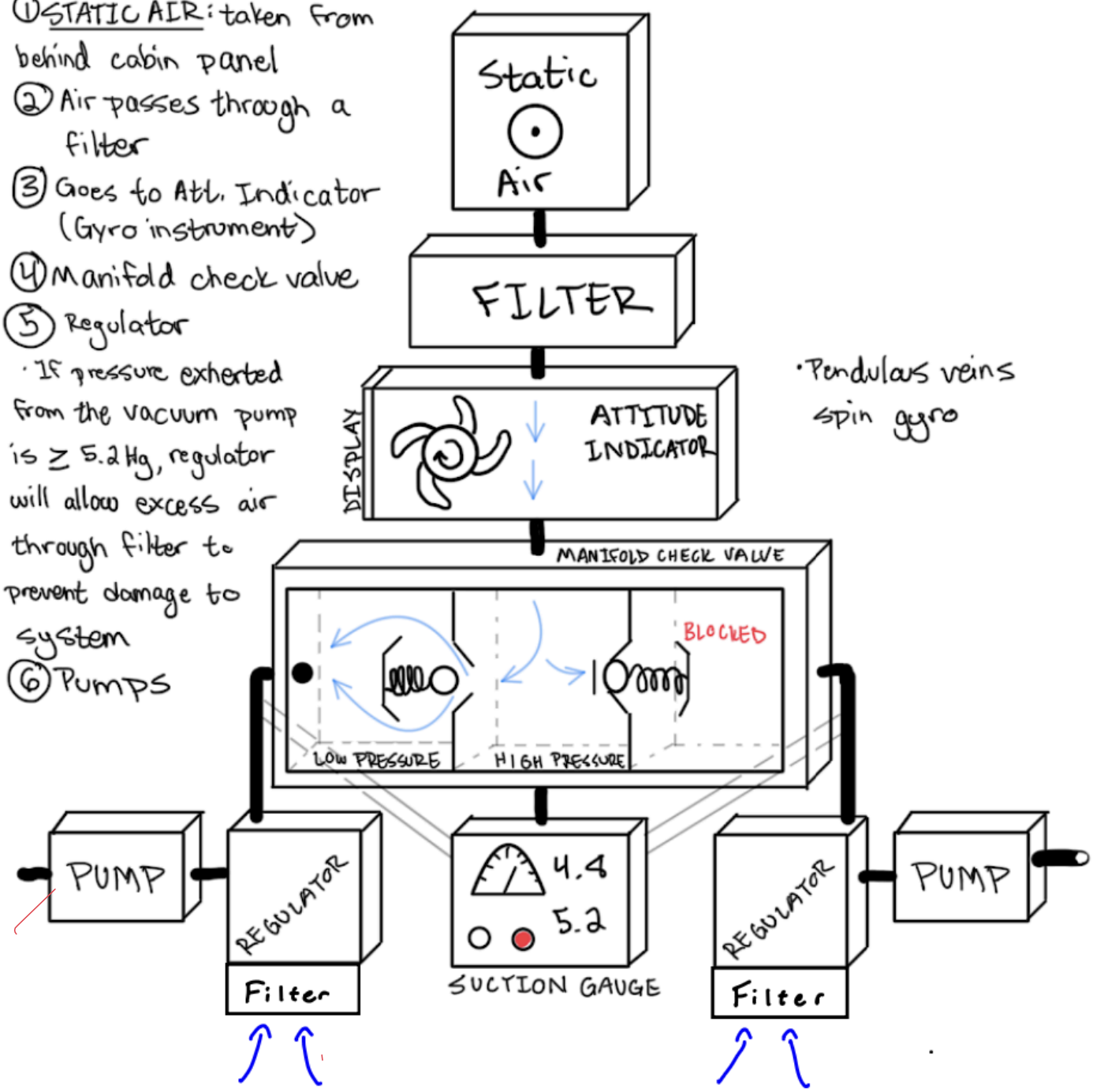
Torque will help counteract Asymetric thrust.

Notes: Controllability and Vmc are always opposite - Lower Vmc = Greater controllability while Higher Vmc = Less controllability
All parameter changes result in lower Vmc (better controllability) and reduced performance, EXCEPT:
Feathering the critical propeller improves both; and wings level makes both worse

ACRONYM	Parameter	Change	Performance	Controllability	Vmc	Explanation
S	Standard Day	Density Altitude Increase				With increase in DA, power of "good" engine decreases, so power difference between engines decreases. Less power difference = Greater Controllability. Less power overall = Less Performance
M	Max Power	Power Reduction				If power of good engine decreases, power difference between engines decreases. Less power difference = Greater controllability. Less power overall = Less performance
A	Aft CG	CG Moves Forward				As CG moves forward, there is a longer arm from rudder to CG, so controllability increases. But more weight forward means that horizontal stabilizer must exert greater down force, so performance decreases (effectively, greater total weight).
C	Critical Engine Windmilling	Critical Engine Feathered				Feathering the critical engine reduces its drag, equivalent to increasing its thrust, thus reducing difference between engines, so controllability increases. Less total drag means a boost to performance.
F	Flaps Up/ Gear Up	Flaps Down/ Gear Down				FLAPS = more overall drag is like less overall thrust = same effects as reducing power. GEAR - lower CG plus "keel effect" increases stability, so increases controllability. All drag decreases performance.
U	Up to 5° Bank	Wings Level				Performance is optimized with no sideslip, so any roll position greater OR less than 5 degrees will impact performance. Controllability benefits from the horizontal component of lift toward the good engine, so more roll is better, less is worse.
M	Most Unfavorable Weight	Add Weight				Adding weight is like decreasing available climb power, but the only engine producing power is the good one, so adding weight is like decreasing the good engine's power, which benefits controllability and impacts performance.

Vacuum System

① STATIC AIR: taken from behind cabin panel
② Air passes through a filter
③ Goes to Atl. Indicator (Gyro instrument)
④ Manifold check valve
⑤ Regulator
· If pressure exherted from the vacuum pump is ≥ 5.2 Hg, regulator will allow excess air through filter to prevent damage to system
⑥ Pumps



· Pendulous veins spin gyro

Bleed Line - for maintenance use, to relieve trapped air in the system

Thermal Relief - routes back to the reservoir as temp. changes in the air

Gear unsafe light - activates when the gear is in transit and will stay illuminated until all 3 limit switches are depressed for specific condition

Gear warning horn:
1. MP is below 14" on 1 or both engines and gear is not down and locked
2. Flaps at 25 or 40 and gear is not down and locked
3. Gear selector is up when on the ground

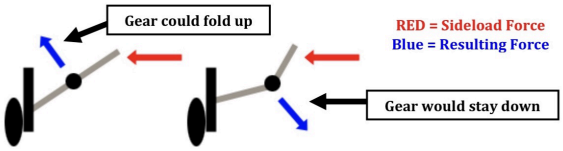
Left squat switch - prevents gear retraction on the ground by disabling the hydraulic pump

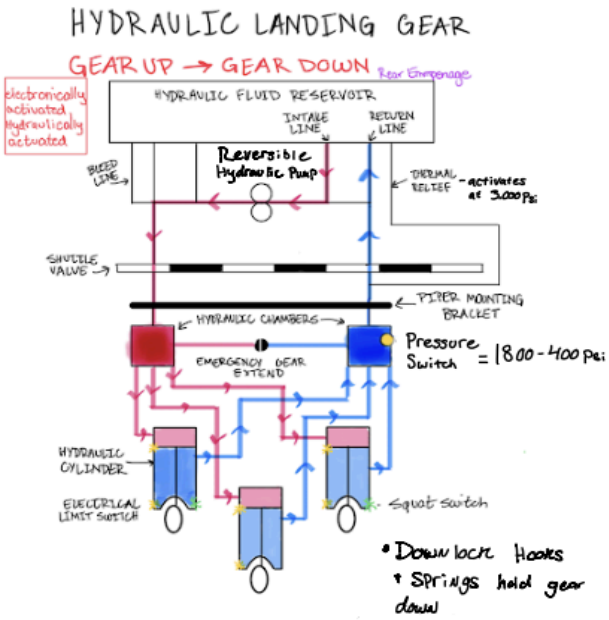
Right squat switch - stall horn & Hobbs time (off/on)

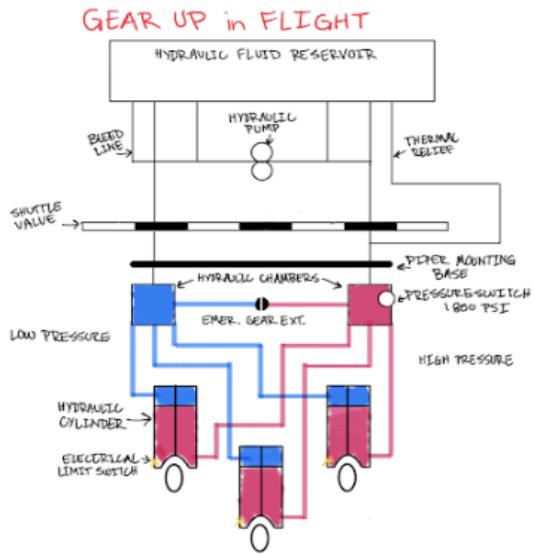
In the up position, the hydraulic pump is shut off by the pressure switch

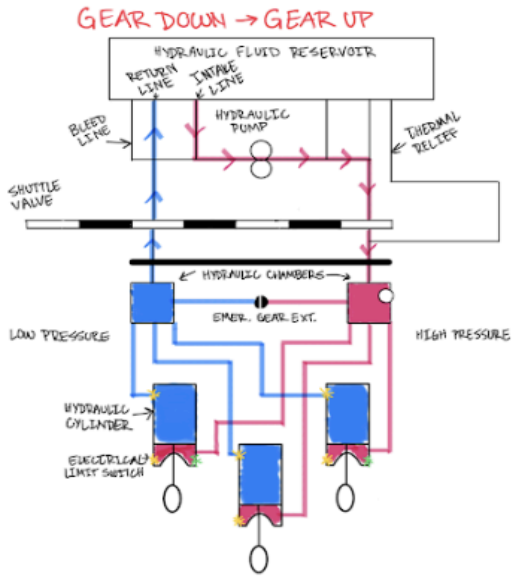
In the down position, the hydraulic pump is shut off by the down limit switches

Over-center joint - helps to keep the gear extended in the event of a side loaded landing



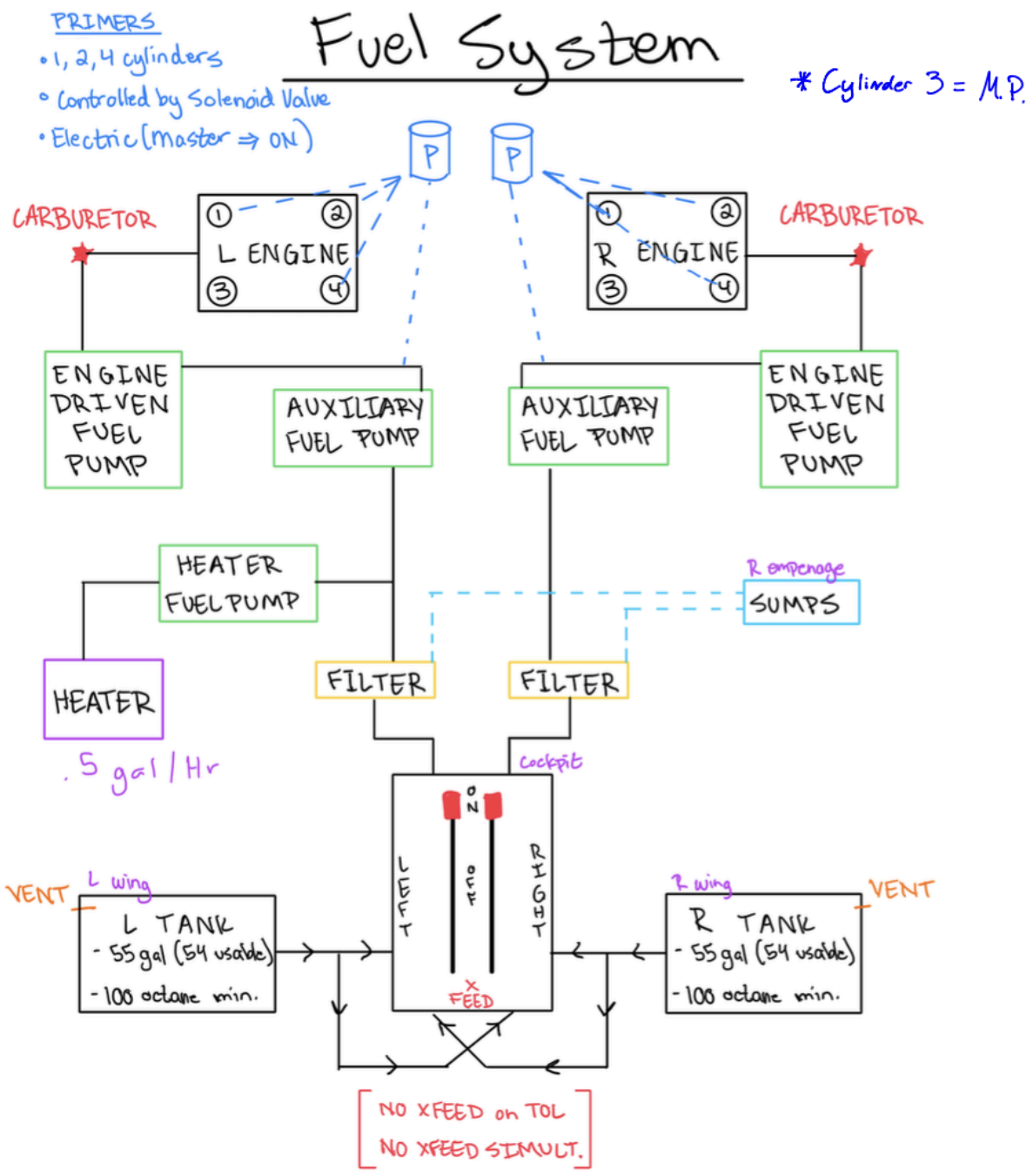






Electrical Limit Switches: When fully extended or retracted, they stop flow of current to motor of hydraulic pump. (6 total, 2 on each gear)

Squat Switch: 79: Located on left main
2000's: Located on both mains



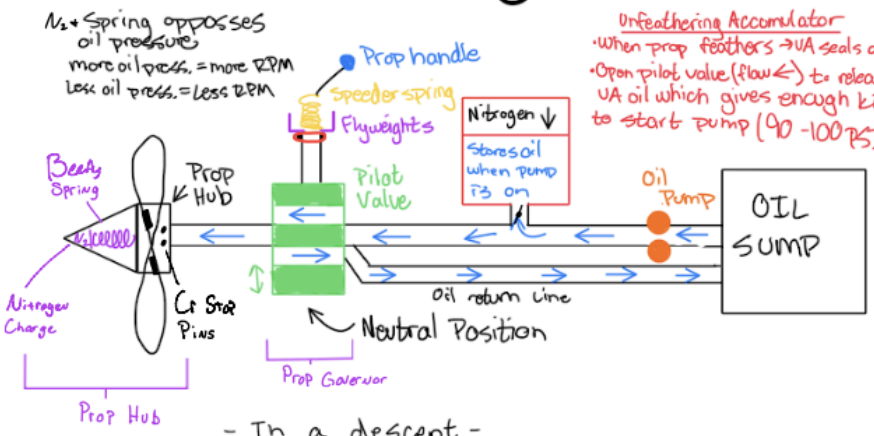
Nacelle rubber bladder tanks

Scupper drain - collects fuel spilled around the bladder

X Feed = engine not fuel tank
Example: Left X Feed - Left engine is supplied fuel from right tank

4 Fuel Vents - 1 under each cap, 1 under each wing

Prop System



- In a descent -

- **Overspeed condition**: RPM's to fast/to much oil
  - Prop speed directly correlates w/ flyweight speed
  - Flyweights move outward lifting pilot valve into oil return position



- In a climb -

- **Underspeed condition**: Slow RPM's/not enough oil
  - Prop + flyweight moving to slow
  - Flyweights move inward pushing down on pilot valve adding more oil

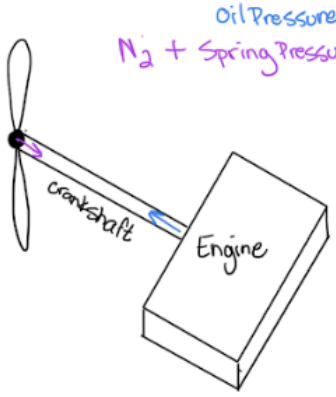


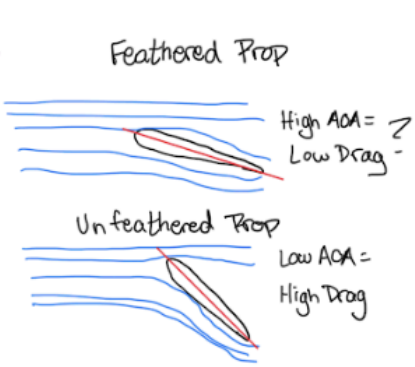
No Unfeathering Accumulator on 79' models, airstart sequence requires use of starter

Prop Hub (AC 65-12(A) pg. 340)

(AC 65-12(A) pg. 350) → Flyweight

more oil press. → more RPM
less oil press → less RPM





Centrifugal Stop Pins

- A feathering lock, operated by centrifugal force
- Prevents feathering when RPM's are below 950

feathering time is 10-17 seconds
unfeathering time is 8-12 seconds ← Depends on oil temp

① Prop Lever



= Feathered

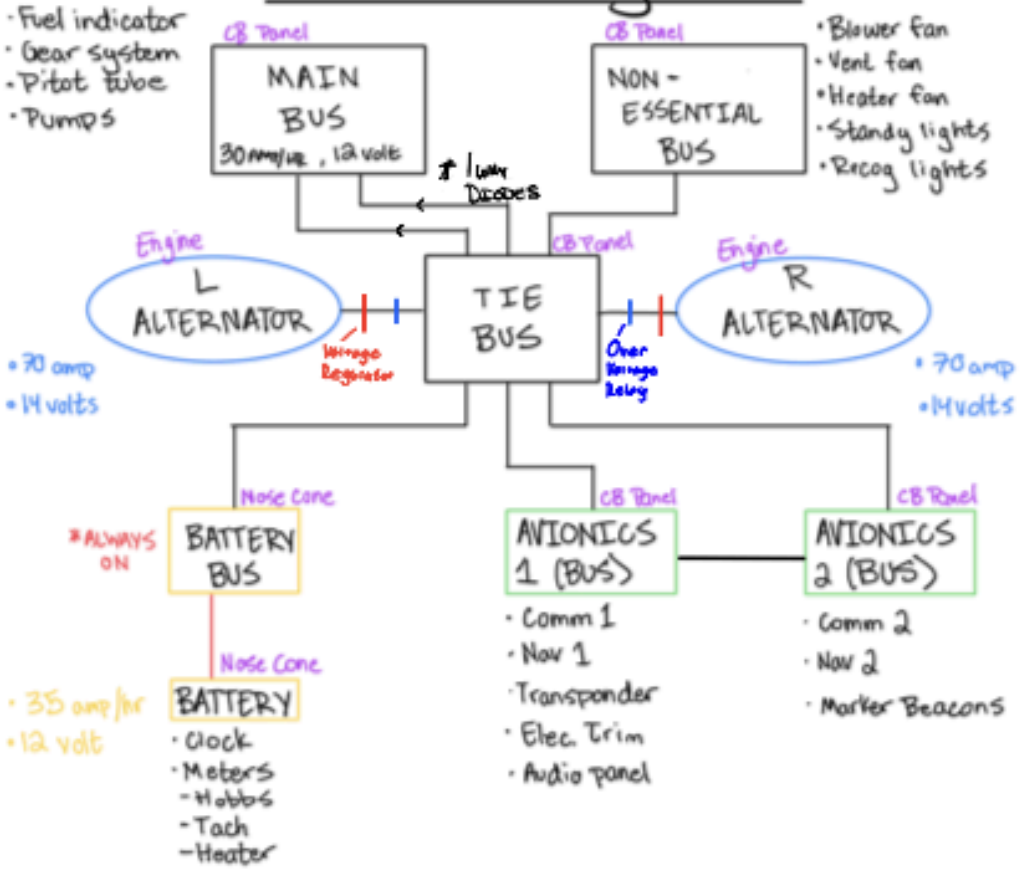
② Oil exits Prop Hub



③ Flyweights return to Neutral, Pilot valve closes, RPM's Set.



Electrical System



Heater System

• Heated air for cabin and defrost is provided by Janitrol Combustion Heater located in forward fuselage. Air moves through manifold and floor ducts to front seats. Heated air from manifold also moves through two ducts to defroster outlets